

(1001CJA102120070)

Test Pattern



CLASSROOM CONTACT PROGRAMME
(Academic Session : 2020 - 2021)

JEE(Main)
FULL SYLLABUS
22-04-2021

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	B	C	D	B	A	D	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	C	C	B	D	A	B	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6.61	11.00	1.46	7.60	80.00	15.00	49.00	2.60	2.00	189.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	C	D	D	B	C	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	D	A	D	B	A	B	B	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6.00	6.00	4.00	3.00	4.00	196.00	2.00	4.00	570.00	60.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	C	D	C	C	B	C	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	C	D	B	D	C	D	A	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3.20	2.00	2010.00	99.99	2.00	8.00	0.00	0.50	0.30	2.00



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PART-1 : PHYSICS

SOLUTION

SECTION-I

1. **Ans. (C)**

$$\text{Sol. } 1(M_1 L_1 T_1^{-2}) = n(M_2 L_2 T_2^{-2})$$

$$n = \left(\frac{M_1}{M_2}\right) \left(\frac{L_1}{L_2}\right) \left(\frac{T_1}{T_2}\right)^{-2}$$

$$= 70 \times 1000 \times (10)^{-2}$$

$$= 700$$

2. **Ans. (A)**

$$\text{Sol. } 20(12) + \frac{1}{2}(a)(12)^2 = 305$$

3. **Ans. (B)**

$$\text{Sol. } a_1 = 2\text{m/s}^2$$

$$a_2 = .75 \text{ m/s}^2$$

$$a_{\text{ret}} = 1.25 \text{ m/s}^2$$

$$\frac{1}{2} a_{\text{ret}} t^2 = 3$$

4. **Ans. (C)**

$$\text{Sol. } -F(5) = -30$$

$$F = 6$$

$$V^2 = 2\left(\frac{3}{4}\right)(8)$$

5. **Ans. (D)**

$$\text{Sol. } P_0^2 = P_x^2 + P_y^2$$

$$36 = 16 + P_y^2$$

$$P_y^2 = 20 \Rightarrow P_y = 2\sqrt{5}$$

$$\tan \theta = \frac{2\sqrt{5}}{4} = \frac{\sqrt{5}}{2}$$

6. **Ans. (B)**

Sol. The final potential of spheres will be

$$\frac{V}{2}, \frac{V}{4}, \frac{V}{8}, \frac{V}{16} \dots$$

Final energy

$$U = \frac{1}{2} 4\pi \epsilon_0 R \left[\left(\frac{V}{2}\right)^2 + \left(\frac{V}{4}\right)^2 + \left(\frac{V}{8}\right)^2 + \left(\frac{V}{16}\right)^2 + \dots \right]$$

$$= \frac{1}{2} 4\pi \epsilon_0 R V^2 \left(\frac{1/4}{1-1/4} \right) = \frac{1}{6} 4\pi \epsilon_0 R V^2$$

Work done by the cell

$$= QV = 4\pi \epsilon_0 R V^2$$

Hence, loss in energy of the system

$$\Delta U = 4\pi \epsilon_0 R V^2 - \frac{1}{6} 4\pi \epsilon_0 R V^2$$

$$= \frac{5}{6} 4\pi \epsilon_0 R V^2$$

$$= \frac{10}{3} \pi \epsilon_0 R V^2$$

7. **Ans. (A)**

$$\text{Sol. } \frac{2\pi r}{T} = \sqrt{\frac{GM}{r}} \Rightarrow r^{3/2} \propto M^{1/2} T$$

$$M^{1/2} \propto \frac{r^{3/2}}{T} \Rightarrow \frac{M_1}{M_2} = \left(\frac{r_1}{r_2}\right)^3 \times \left(\frac{T_2}{T_1}\right)^2$$

$$= \left(\frac{1}{3}\right)^3 \times \left(\frac{3}{2}\right)^2 = \frac{1}{27} \times \frac{9}{4} = 12$$

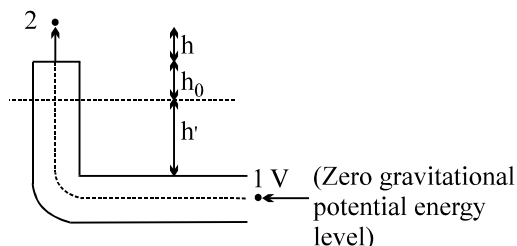
$$M_2 = 12M_1$$

8. **Ans. (D)**

$$\text{Sol. } \frac{4}{R} \times 36 = \frac{\epsilon}{R} \times 48 \Rightarrow \epsilon = 3 \text{ volt}$$

9. Ans. (B)

Sol. Applying Bernoulli's theorem for points 1 and 2 of the streamline shown
 $P_0 \rightarrow$ Atmospheric pressure



$$\frac{1}{2} \rho V^2 + (P_0 + \rho g h') + 0$$

$$= 0 + P_0 + \rho g (h' + h_0 + h)$$

$$\Rightarrow h = 10 \text{ cm}$$

10. Ans. (B)

Sol. $m = \frac{f_0}{f_e} = \frac{20}{2.5 \times 10^{-2}} = 800$

11. Ans. (D)

Sol. $v' - v = 4$

$$v \left(\frac{v}{v - v_s} - 1 \right) = 4$$

$$110 \left(\frac{v_s}{v - v_s} \right) = 1$$

$$110v_s = v - v_s$$

$$111v_s = 333$$

$$v_s = 3 \text{ m/s}$$

12. Ans. (C)

Sol. Average $v_s = 6$

$$\text{Reading} = 0.50 + 0.01 \times 6$$

$$\text{Corrected reading} = 0.56 + (0.03)$$

13. Ans. (C)

Sol. $\frac{\mu_0 i}{2R} \frac{(2\pi - \theta)}{2\pi} + \frac{\mu_0 i \theta}{2r 2\pi} \dots (1)$

$$\frac{\mu_0 i}{2R} \frac{(2\pi - \theta)}{2\pi} - \frac{\mu_0 i \theta}{2r 2\pi} \dots (2)$$

$$\frac{\mu_0 i}{2r} \frac{2\pi - \theta}{2\pi} + \frac{\mu_0 i}{2R} \frac{\theta}{2\pi} \dots (3)$$

$$\frac{2\pi - \theta}{R} + \frac{\theta}{r} = \frac{2\pi}{R} - \frac{\theta}{R} + \frac{\theta}{r} \dots (1)$$

$$-\frac{\theta}{R} + \frac{\theta}{r} \dots (2)$$

$$+\frac{2\pi}{r} - \frac{\theta}{r} + \frac{\theta}{R} \dots (3)$$

14. Ans. (C)

Sol. $I_1 + I_2 = 3I_0$

$$(\sqrt{I_1} - \sqrt{I_2})^2 = I_0$$

$$I_1 + I_2 - 2\sqrt{I_1} \sqrt{I_2} = I_0$$

$$2\sqrt{I_1} \sqrt{I_2} = 2I_0$$

$$I_{\max} = 3I_0 + 2I_0 = 5I_0$$

15. Ans. (B)

Sol. $F = Bi\ell = B\ell \frac{B_0 \ell}{R}$

$$F_1 : F_2 : F_3 : F_4 = \frac{4\ell^2}{6\ell} : \frac{4\ell^2}{8\ell} : \frac{\ell^2}{6\ell} : \frac{\ell^2}{4\ell}$$

16. Ans. (D)

Sol. $X_L = \frac{1}{2} \times 1000 = 500 \Omega$

$$X_C = \frac{10^6}{5 \times 1000} = 200 \Omega$$

$$z = \sqrt{400^2 + 300^2}$$

$$z = 500 \Omega$$

$$i_{\max} = \frac{1}{5} \text{ A}$$

$$P = i_{\text{rms}}^2 R = \frac{400}{2 \times 25} = 8 \text{ W}$$

17. Ans. (A)

Sol. Distance between two position of bridge is $\lambda/2$ (Distance between two nodes)

$$\frac{\lambda}{2} = 25\text{cm}$$

$$\lambda = 50\text{ cm}$$

$$v = n\lambda \Rightarrow \sqrt{\frac{T}{\mu}} = n\lambda$$

$$n = 400; T = 20\text{ N}$$

$$\mu = 0.5\text{ gm/m}$$

18. Ans. (B)

Sol. AB: $\Delta U = Q + W$, but

[w = work done on the gas]

$W = 0$, so $\Delta U = Q$.

Temperature decreases from A to B, so ΔU must be negative, therefore Q must be negative. This implies that heat is taken from the gas.

BC: The gas is expanding so W is negative, and temperature is increasing, so ΔU is +ve while W is negative, therefore Q must be positive, therefore heat is added to the gas.

CD: $\Delta U = Q + W$, but $W = 0$, so $\Delta U = Q$.

Temperature increases from C to D so Q is positive, therefore heat is added to the gas.

DA: ΔU is zero because DA is an isotherm, therefore $Q = -W$. The gas is compressed, so W is positive, therefore Q must be negative, indicating that heat is taken from the gas.

19. Ans. (A)

Sol. $P = eA\sigma T^4$

$$\frac{P_1}{P_2} = \frac{1}{1} \times \frac{6^2}{18^2} \times \left(\frac{T_1}{T_2}\right)^4$$

$$\frac{P_1}{P_2} = \frac{1}{9} \times 3^4 = 9$$

$$T_1\lambda_1 = T_2\lambda_2$$

$$\frac{T_1}{T_2} = \frac{1500}{500} = 3$$

20. Ans. (A)

Sol. Let initially the number of nuclei of $^{32}_{15}\text{P}$ & $^{33}_{15}\text{P}$ are N_1 & N_2 respectively

$$\text{so } \frac{R_1}{R_2} = \frac{\lambda_1 N_1}{\lambda_2 N_2} = \frac{90}{10} \quad (\because 90\% \text{ from } ^{32}_{15}\text{P} \text{ \& } 10\% \text{ from } ^{33}_{15}\text{P})$$

Now let at time 't' 90% decays come from $^{33}_{15}\text{P}$ then ratio of activity.

$\frac{R_1'}{R_2'} = \frac{\lambda_1 N_1'}{\lambda_2 N_2'} = \frac{\lambda_1 N_1 e^{-\lambda_1 t}}{\lambda_2 N_2 e^{-\lambda_2 t}} = \frac{10}{90} \Rightarrow$

$$\frac{R_1'}{R_2'} = \frac{\lambda_1 N_1'}{\lambda_2 N_2'} \Rightarrow \frac{R_1'}{R_2'} = \frac{\lambda_1 N_1 e^{-\lambda_1 t}}{\lambda_2 N_2 e^{-\lambda_2 t}} = \frac{10}{90} \Rightarrow$$

$$\frac{90e^{-\lambda_1 t}}{10e^{-\lambda_2 t}} = \frac{10}{90}$$

$$\Rightarrow e^{(\lambda_1 - \lambda_2)t} = 81 \Rightarrow \ell n e^{(\lambda_1 - \lambda_2)t} = \ell n 3^4$$

$$\Rightarrow (\lambda_1 - \lambda_2)t = 4\ell n 3 \Rightarrow t = \frac{4\ell n 3}{\ell n 2 \left(\frac{1}{T_1} - \frac{1}{T_2} \right)}$$

SECTION-II

1. Ans. 6.61 A°

$$\text{Sol. } 2\pi r = 5 \frac{h}{p} \Rightarrow 2\pi r = 5 \frac{h}{p}$$

$$n = 5$$

$$r = 0.529 \times \frac{25}{2} = 6.61\text{ A}^\circ$$

2. Ans. 11.00

$$\text{Sol. } = \frac{m\ell^2}{4} + \left(\frac{m\ell^2}{12} \right) 2 + 2 \left(\frac{m\ell^2}{4} \right)$$

$$= m\ell^2 \left(\frac{1}{4} + \frac{1}{6} + \frac{1}{2} \right)$$

$$= m\ell^2 \left(\frac{3+2+6}{12} \right) = \frac{11}{12} m\ell^2$$

3. Ans. 1.46

Sol. $3F_1 + F_2 = 3900$

$$F = \frac{AY}{\ell} x$$

$$3 \frac{AY}{\ell} x + \frac{AY}{\ell} (x + 0.5 \times 10^{-3}) = 3900$$

$$4x + 0.5 \times 10^{-3} = 3 \times 10^{-3}$$

$$4x = 2.5 \times 10^{-3}$$

$$x = \frac{2.5}{4} \times 10^{-3}$$

$$F_2 = 1.3 \times 10^6 \left(\frac{2.5}{4} + 0.5 \right) 10^{-3}$$

$$= 1.3 \times 10^3 \left(\frac{4.5}{4} \right)$$

$$F_2 = 1.46 \times 10^3 = 146 \text{ N}$$

4. Ans. 7.60

Sol. $\beta = \frac{\alpha}{1-\alpha} = \frac{0.95}{1.095} = 19$

$$\beta = \frac{\Delta I_c}{\Delta I_b}$$

$$\Delta I_c = \beta \Delta I_b$$

$$= 19 \times 0.95$$

$$= 7.6 \text{ mA}$$

5. Ans. 80.00

Sol. $d = \sqrt{2Rh}$

$$d = \sqrt{2 \times 6400 \times \frac{1}{2}} = 80 \text{ km}$$

6. Ans. 15.00

Sol. $76 = 95 \left(1 - e^{-\frac{t}{RC}} \right)$

$$e^{-\frac{t}{RC}} = 1 - \frac{76}{95} = 5^{-1}$$

$$\frac{t}{RC} = \ln 5$$

$$\frac{1}{2(\ln 5)} = RC$$

$$\frac{1}{2 \ln 5} = \frac{100}{3 \ln 5} \times 10^{-6} \text{ R}$$

7. Ans. 49.00

Sol. $f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$

8. Ans. 2.60

Sol. $d \sin \theta = \lambda$

$$d \times \sin 15 = 650 \times 10^{-9}$$

$$d = \frac{650 \times 10^{-9}}{0.25}$$

$$d = 2600 \times 10^{-9}$$

$$d = 2.600 \mu\text{m}$$

9. Ans. 2.00

Sol. $5 = \frac{v^2}{R}$

$$(\sqrt{5R})^3 = \frac{3\pi}{2} \times R$$

$$5R = \frac{10}{4} \times R^2$$

$$R = \frac{20}{10} = 2 \text{ m}$$

10. Ans. 189.00

Sol. $B = \mu_r \mu_0 n i$

$$B = 100 \times 4\pi \times 10^{-7} \times \frac{630}{2\pi \times 20 \times 10^{-2}} \times 3$$

$$= \frac{100 \times 2 \times 10^{-5} \times 630}{20} \times 3$$

$$B = 189 \times 10^{-3}$$

$$= 189 \text{ mT}$$

SECTION-I

1. Ans. (C)

2. Ans. (D)

Sol. According to Arrhenius equation

$$k = A \cdot e^{-E_a/RT}$$

In presence of catalyst

$$k' = A \cdot e^{-E_a'/RT}$$

$$\text{Given } E_a' = 4.15 \text{ kJ mol}^{-1}$$

..... in the presence of a catalyst

$$\therefore \frac{k'}{k} = e^{(E_a - E_a')/RT}$$

$$\text{Given } k' = (1 + 1.718) k = 2.718k = ek$$

$$\text{Hence } e = e^{(E_a - E_a')/RT}$$

$$\text{or } E_a - E_a' = RT$$

$$\text{or } E_a = E_a' + RT = 4.15 + 8.3 \times 500 \times 10^{-3} = 8.3 \text{ kJ mol}^{-1}$$

3. Ans. (C)

4. Ans. (C)

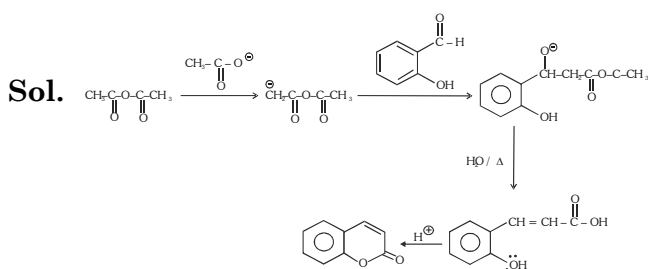
5. Ans. (D)

6. Ans. (D)

7. Ans. (B)

8. Ans. (C)

9. Ans. (A)

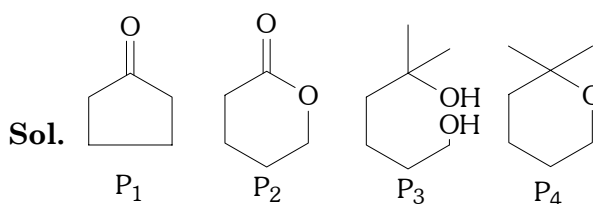


10. Ans. (B)

Sol. β, γ -Unsaturated carboxylic acid undergoes decarboxylation only on heating.

11. Ans. (B)

12. Ans. (A)



13. Ans. (D)

14. Ans. (A)

15. Ans. (D)

16. Ans. (B)

$$\text{Sol. } \frac{t_{\text{effuse}} \text{CH}_4}{t_{\text{effuse}} \text{He}} = \sqrt{\frac{m_{\text{CH}_4}}{m_{\text{He}}}}$$

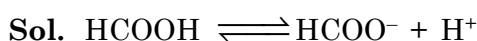
$$\frac{48}{T} = \sqrt{\frac{16}{4}}$$

17. Ans. (A)

18. Ans. (B)

19. Ans. (B)

20. Ans. (B)



$$0.1 - X \quad X \quad X$$

$$\Delta T_f = 0.2046 = 1.86 \times m$$

$$m = \frac{0.2046}{1.86} = 0.1 + X = 0.11$$

$$K_c = \frac{K_w}{K_a}$$

$$\therefore K_a = \frac{X^2}{0.1 - X} = \frac{10^{-2}}{9}$$

$$\Rightarrow K_b (\text{HCOO}^-) = 9 \times 10^{-12} \text{ M}$$

SECTION-II

1. Ans. (6.00)

2. Ans. (6.00)

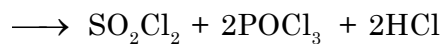
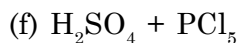
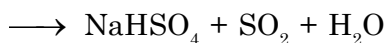
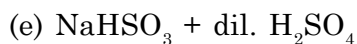
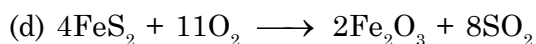
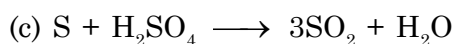
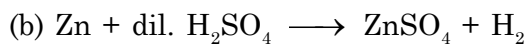
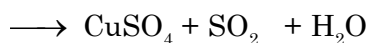
3. Ans. (4.00)

Sol. $[\text{AgNO}_3, \text{Pb}(\text{OAc})_2, \text{HgCl}_2, \text{CuSO}_4]$

4. Ans. (3.00)

5. Ans. (4.00)

Sol. (a) $\text{Cu} + \text{conc. H}_2\text{SO}_4$



6. Ans. (196.00)

Sol. $\text{Ag}_2\text{A} \xrightarrow{\Delta} \text{Ag}$

0.41 0.216

POAC on Ag

$$= 2 \left(\frac{0.41}{2 \times 108 + \text{MM}_{\text{A}^{2-}}} \right) = \frac{0.216}{108} \Rightarrow \text{MM}_{\text{A}^{2-}} = 194$$

$$\therefore \text{MM}_{\text{H}_2\text{A}} = 2 \times 1 + \text{MM}_{\text{A}^{2-}}$$

$$= 2 + 194$$

$$= 196 \text{ gm/mole}$$

7. Ans. (2.00)

Sol. $[\text{OH}^-]_{\min}$ to ppt

$$\text{Fe}(\text{OH})_2 = \left(\frac{8 \times 10^{-16}}{0.02} \right)^{\frac{1}{2}} = 2 \times 10^{-7}$$

$$\therefore \text{pH}_{\min} = 7.3$$

$[\text{OH}^-]_{\min}$ to ppt

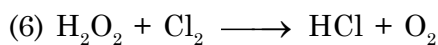
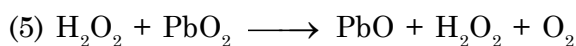
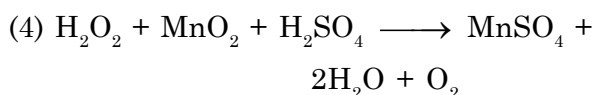
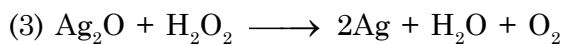
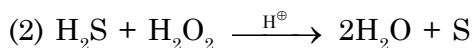
$$\text{Fe}(\text{OH})_3 = \left(\frac{4 \times 10^{-28}}{0.05} \right)^{\frac{1}{3}} = 2 \times 10^{-9}$$

$$\therefore \text{pH}_{\min} = 5.3$$

$$\therefore \Delta \text{pH} = 7.3 - 5.3 = 2.0 \text{ unit}$$

8. Ans. (4.00)

Sol. (1) $2\text{KI} + \text{H}_2\text{O}_2 \xrightarrow{\text{H}^+} 2\text{KOH} + \text{I}_2$



9. Ans. (570.00)

10. Ans. (60.00)

SECTION-I

1. Ans. (A)

Sol. $\frac{1}{\alpha\beta} = \frac{b}{a} + \frac{4a}{b} + 1$

as $\frac{b}{a} + \frac{4a}{b} + 1 \geq 5$

so $\left(\frac{10}{5 + \alpha\beta}\right)_{\max} = 1$

2. Ans. (A)

Sol. $x^2 f(x) - 2f\left(\frac{1}{x}\right) = g(x)$

and $2f\left(\frac{1}{x}\right) - 4x^2 f(x) = 2x^2 g\left(\frac{1}{x}\right)$

$\therefore -3x^2 f(x) = g(x) + 2x^2 g\left(\frac{1}{x}\right)$

$\therefore f(x) = -\left[\frac{g(x) + 2x^2 g\left(\frac{1}{x}\right)}{3x^2}\right]$

$\therefore g(x)$ and x^2 are odd and even function respectively. So, $f(x)$ is an odd function.

But $f(x)$ is given even.

$\therefore f(x) = 0 \forall x$.

Hence, $f(5) = 0$

3. Ans. (C)

Sol. Given that $f(x) = |1 - x|$

$\Rightarrow f(|x|) = \begin{cases} x-1, & x > 1 \\ 1-x, & 0 < x \leq 1 \\ 1+x, & -1 \leq x \leq 0 \\ -x-1, & x < -1 \end{cases}$

Clearly the domain of $\sin^{-1}(f(|x|))$ is $[-2, 2]$.

$\Rightarrow$ It is non differentiable at the points

$\{-1, 0, 1\}$.

4. Ans. (D)

Sol. Since $y = \int_0^x f(t) \sin\{k(x-t)\} dt \dots (i)$

$\therefore \frac{dy}{dx} = \int_0^x \frac{\partial}{\partial x} [f(t) \sin k(x-t)] dt + \frac{dx}{dt} [f(x) \sin k(x-x)]$

$-\frac{d}{dx}(0) \{f(0) \cdot \sin(k(x-0))\}$

$= k \int_0^x f(t) \cos(k(x-t)) dt$

Again differentiating both w.r.t $x \frac{d^2 y}{dx^2} = k$

$\left[\int_0^x \frac{\partial}{\partial x} f(t) \cos(k(x-t)) dt + \frac{dx}{dx} \right]$

$\{f(x) \cos k(x-0)\} - \frac{d}{dx} 0 \{f(0) \cos k(x-0)\}$

$= k \left[-k \int_0^x f(t) \sin(k(x-t)) dt + f(x) - 0 \right]$

$= -k^2 y + k f(x) \quad [\text{from (1)}]$

5. Ans. (C)

Sol. ${}^{(100-55)+(10-1)}C_{10-1} = {}^{54}C_9$

6. Ans. (C)

Sol. $\sum_{n=1}^5 \sin^{-1} \left(\sin \left(2n\pi - \frac{\pi}{6} \right) \right) + \sum_{n=1}^5 \sin^{-1} \left(\sin (2n-1)\pi - \frac{\pi}{6} \right)$

$\sum_{n=1}^5 \sin^{-1} \left(-\sin \frac{\pi}{6} \right) + \sum_{n=1}^5 \sin^{-1} \left(\sin \frac{\pi}{6} \right)$

$= -\frac{5\pi}{6} + \frac{5\pi}{6} = 0$

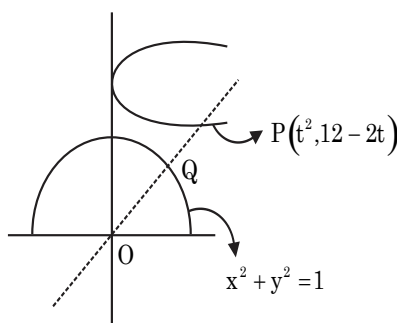
7. Ans. (B)

Sol. $A(x_1, \sqrt{1-x_1^2})$

$B(x_2, 12-2\sqrt{x_2})$

Consider two curves:

$$y = \sqrt{1-x^2} \text{ \& } y = 12 - 2\sqrt{x}$$



Minimum distance occurs along common normal

$\Rightarrow$ Normal at P must pass through (0, 0)

$$y = 12 - 2\sqrt{x}$$

$$P \rightarrow (4, 8)$$

$$\therefore OP = \sqrt{16 + 64} = \sqrt{80}$$

$$PQ = OP - OQ$$

$$= \sqrt{80} - 1$$

8. **Ans. (C)**

Sol. vector $(2\hat{i} - 3\hat{j}) \times (\hat{i} - 4\hat{j} + 4\hat{k})$ is perpendicular to $2\hat{i} - \hat{j} + m\hat{k}$

$$\Rightarrow \begin{vmatrix} 2 & -3 & 0 \\ 1 & -4 & 4 \\ 2 & -1 & m \end{vmatrix} = 0 \Rightarrow m = \frac{-16}{5}$$

9. **Ans. (D)**

Sol. $(p \vee \sim r) \vee (\sim p \wedge q \wedge r)$

$$= (p \vee \sim r) \vee (\sim(p \vee \sim r) \wedge q)$$

$$= (p \vee \sim r) \vee q = \sim(\sim p \wedge \sim q \wedge r)$$

10. **Ans. (A)**

Sol. $3x^2 \frac{dx}{dy} = \frac{x^3}{x^3 - y} \quad x^3 = t$

$$\Rightarrow \frac{dt}{dy} = \frac{t}{t-y} \Rightarrow (t-y)dt = t dy$$

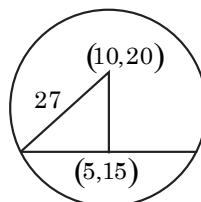
$$\Rightarrow t dt = y dt + t dy \Rightarrow \frac{t^2}{2} = t.y + \frac{c}{2}$$

$$\Rightarrow x^6 = 2x^3 y + c$$

11. **Ans. (D)**

Sol. Clearly (5,15) is the mid point of the chord hence slope of normal to the chord is

$$\frac{20-15}{10-5} = 1, \text{ slope of chord} = -1$$



$$\text{Equation of chord } (y-15) = -(x-5)$$

$$\Rightarrow x + y = 20$$

$$\Rightarrow 2x + 2y = 40$$

$$\Rightarrow p^{2013} + q^{2013} = (2)^{2013} + (2)^{2013} = 2^{2014}$$

12. **Ans. (C)**

Sol. $D = (2-a)(3-b)$

$$Dx = (4b-15)(a-2)$$

$$Dy = 0, Dz = a-2$$

If $a \neq 2$ and $b = 3$ then $D = 0, Dx \neq 0, Dz \neq 0$

$\therefore$ Given system of equation has no solution

$\therefore$ Option (C) is incorrect.

13. **Ans. (C)**

Sol. $2, H_1, H_2, \dots, H_{20}, 3$ are in H.P.

$$\frac{1}{2}, \frac{1}{H_1}, \frac{1}{H_2}, \dots, \frac{1}{H_{20}}, \frac{1}{3} \text{ are in A.P.}$$

Suppose common difference is d

$$\frac{1}{2} + \frac{1}{H_1} = \frac{1}{3} + \frac{1}{H_{20}}$$

Now $\frac{1}{2} - \frac{1}{H_1} = \frac{1}{3} - \frac{1}{H_{20}}$

$$\begin{aligned} \frac{1}{2} + \frac{1}{H_1} &= \frac{1}{3} + \frac{1}{H_{20}} \\ \frac{1}{-d} + \frac{1}{d} &= \frac{1}{3} - \frac{1}{2} + \frac{1}{H_{20}} - \frac{1}{H_1} \\ &= \frac{21d + 19d}{d} = 40 \end{aligned}$$

14. Ans. (D)

Sol. $\lim_{x \rightarrow 1} \frac{f'(x)}{2} = \frac{\pi - 2}{16\sqrt{2}}$

15. Ans. (B)

Sol. $\left(y + \frac{2}{3}\right)^2 = 2\left(x - \frac{10}{9}\right)$

$$Y^2 = 2X$$

$$X > 2a, Y = 0$$

$$X > \frac{19}{9}, Y = -\frac{2}{3}$$

16. Ans. (D)

Sol. Plane containing the line of intersection of the given planes is

$$(x - y - z - 4) + \lambda(x + y + 2z - 4) = 0$$

s.e.

$$(\lambda + 1)x + (\lambda - 1)y + (2\lambda - 1)z - 4(\lambda + 1) = 0 \dots (i)$$

Vector normal to it

$$\vec{V} = (\lambda + 1)\hat{i} + (\lambda - 1)\hat{j} + (2\lambda - 1)\hat{k}$$

Vector along the line of intersection of the planes

$$2x + 3y + z - 1 = 0 \text{ \& } x + 3y + 2z - 2 = 0 \text{ is}$$

$$\text{Given by } \vec{n} = 3(\hat{i} - \hat{j} + \hat{k})$$

\& \vec{n} is parallel to the plane $\rightarrow (i)$

$$\Rightarrow \vec{n} \cdot \vec{V} = 0 \Rightarrow \lambda = \frac{-1}{2}$$

\& hence the required plane is

$$x - 3y - 4z - 4 = 0$$

$$\Rightarrow |A + B + C| = 11$$

17. Ans. (C)

Sol. $\tan^{-1} x$ is increasing function

$$\Rightarrow T_n < S_n$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} T_n = \int_0^1 \tan^{-1} x dx = \frac{\pi - \ln 4}{4}$$

$$S_n > \frac{\pi - \ln 4}{4}, T_n < \frac{\pi - \ln 4}{4}$$

18. Ans. (D)

Sol. Reflexive $\forall f \in R, f - f = 0 \in R$

$$\Rightarrow (f, f) \in R$$

Symmetric $\forall (f, g) \in R$

$$\Rightarrow f - g \text{ is even}$$

$$\Rightarrow g - f \text{ is even}$$

$$\Rightarrow (g, f) \in R$$

Transitive $\forall (f, g) \in R, (g, h) \in R$

$$\Rightarrow f - g \text{ is even, } g - h \text{ is even}$$

$$\Rightarrow f - h \text{ is even}$$

$$\Rightarrow (f, h) \in R$$

So, R is equivalence relation

19. Ans. (A)

Sol. $\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{6}\right) \dots \left(1 - \frac{1}{2012}\right)$

$$= \frac{1}{2} \times \frac{3}{4} \times \frac{5}{6} \times \frac{7}{8} \times \dots \times \frac{2011}{2012} = \frac{2012!}{2^{2012} (1006!)^2}$$

20. Ans. (C)

Sol. $x_1, x_2, \dots, x_n$ be n observations then,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{Let } y_i = \frac{x_i}{\alpha} + 10 \text{ then}$$

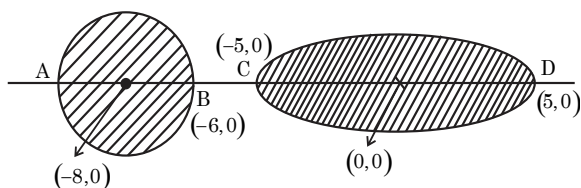
$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i = \frac{1}{\alpha} \left(\frac{1}{n} \sum_{i=1}^n x_i \right) + \frac{1}{n} (10n)$$

$$\Rightarrow \bar{y} = \frac{\bar{x}}{\alpha} + 10 = \frac{\bar{x} + 10\alpha}{\alpha}$$

SECTION-II

1. Ans. 3.20

Sol.



AD is maximum distance BC is minimum distance Range is $[1, 15]$

2. Ans. 2.00

Sol. $f(x) = (x-1)(x-\cos\theta)(x-\sin\theta)$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 + \sin^2\theta + \cos^2\theta = 2$$

3. Ans. 2010.00

Sol. $I_n = \int_0^1 x^n \tan^{-1} x dx$

Integrating by parts

$$I_n = \left(\frac{x^{n+1}}{n+1} \tan^{-1} x \right)_0^1 - \frac{1}{n+1} \int_0^1 \frac{x^{n+1}}{1+x^2} dx$$

$$= \frac{\pi}{4(n+1)} - \frac{1}{n+1} \int_0^1 \frac{x^{n-1}(x^2+1-1)}{x^2+1} dx$$

$$\Rightarrow (n+1)I_n = \frac{\pi}{4} - \int_0^1 x^{n-1} dx + \int_0^1 \frac{x^{n-1}}{x^2+1} dx$$

$$\Rightarrow (n+1)I_n = \frac{\pi}{4} - \frac{1}{n} + \left(x^{n-1} \tan^{-1} x \right)_0^1 - \int_0^1 (n-1)x^{n-2} \tan^{-1} x dx$$

$$= \frac{\pi}{4} - \frac{1}{n} + \frac{\pi}{4} - (n-1) \int_0^1 x^{n-2} \tan^{-1} x dx$$

$$\Rightarrow (n+1)I_n = \frac{\pi}{2} - \frac{1}{n} - (n-1)I_{n-2}$$

$$\Rightarrow (n+1)I_n + (n-1)I_{n-2} = \frac{\pi}{2} - \frac{1}{n}$$

Put $n = 2010$

4. Ans. 99.99

Sol. $\sum_{k=0}^{100} \frac{k+1-1}{k+1} {}^{100}C_k = \sum_{k=0}^{100} {}^{100}C_k - \sum_{k=0}^{100} \frac{{}^{100}C_k}{k+1}$

$$= 2^{100} - \frac{1}{101} \sum_{k=0}^{100} \frac{101}{k+1} {}^{100}C_k$$

$$= \frac{99 \cdot 2^{100} + 1}{101} \quad (x=99, y=1 \text{ \& } z=101)$$

$$\Rightarrow \frac{xy^2z}{100} = 99.99$$

5. Ans. 2.00

Sol. $\Delta = \frac{1}{2} \times 4 \sqrt{\frac{2}{5}} \times \frac{|2\cos\theta - \sin\theta|}{\sqrt{2}} \leq 2$

6. Ans. 8.00

Sol. $|A| = \begin{vmatrix} 1 & \tan x \\ -\tan x & 1 \end{vmatrix} = 1 + \tan^2 x \neq 0$

So A is invertible.

Let C_{ij} be the cofactor of a_{ij} in $A = [a_{ij}]$

Then

$$C_{11} = (-1)^{1+1} 1 = 1, C_{12} = (-1)^{1+2} (-\tan x) = \tan x$$

$$C_{21} = (-1)^{2+1} \tan x = -\tan x, C_{22} = (-1)^{2+2} 1$$

$$\therefore \text{adj } A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

$$\text{Now, } A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$\Rightarrow A^{-1} = \frac{1}{(1 + \tan^2 x)} \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{1 + \tan^2 x} & \frac{-\tan x}{1 + \tan^2 x} \\ \frac{\tan x}{1 + \tan^2 x} & \frac{1}{1 + \tan^2 x} \end{bmatrix}$$

$$\therefore A^T A^{-1} = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{1 + \tan^2 x} & \frac{-\tan x}{1 + \tan^2 x} \\ \frac{\tan x}{1 + \tan^2 x} & \frac{1}{1 + \tan^2 x} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1 - \tan^2 x}{1 + \tan^2 x} & \frac{-2\tan x}{1 + \tan^2 x} \\ \frac{2\tan x}{1 + \tan^2 x} & \frac{1 - \tan^2 x}{1 + \tan^2 x} \end{bmatrix} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$$

7. Ans. 0.00

Sol. Given curve is $x^3 + y^3 = c^3$

$$\therefore \frac{dy}{dx} = -\frac{x^2}{y^2} \Rightarrow \frac{dy}{dx}_{(a,b)} = -\frac{a^2}{b^2} \quad \dots (1)$$

and tangent at (a, b) cuts the curve again

at (a_1, b_1)

$\therefore$ slope of tangent

$$= \frac{b_1 - b}{a_1 - a} = -\frac{a^2}{b^2} \quad \{\text{from (1)}\} \dots (2)$$

$$\text{Also } a^3 + b^3 = c^3 \quad \dots (3)$$

$$\text{and } a_1^3 + b_1^3 = c^3 \quad \dots (4)$$

Subtracting (4) from (3), we get

$$(a^3 - a_1^3) + (b^3 - b_1^3) = 0$$

$$\Rightarrow (a - a_1)(a^2 + aa_1 + a_1^2) + (b - b_1)(b^2 + bb_1 + b_1^2) = 0$$

$$\Rightarrow \frac{b_1 - b}{a_1 - a} = -\frac{a^2 + aa_1 + a_1^2}{b^2 + bb_1 + b_1^2} \quad \dots (5)$$

$$\text{From (2) and (5), } -\frac{a^2}{b^2} = -\frac{a^2 + aa_1 + a_1^2}{b^2 + bb_1 + b_1^2}$$

$$\Rightarrow a^2b^2 + a^2bb_1 + a^2b_1^2 = a^2b^2 + ab^2a_1 + a_1^2b^2$$

$$\Rightarrow ab(ab_1 - ba_1) + a^2b_1^2 - a_1^2b^2 = 0$$

$$\Rightarrow ab(ab_1 - ba_1) + (ab_1 + a_1b)(ab_1 - a_1b) = 0$$

$$\Rightarrow (ab_1 - a_1b)(ab + ab_1 + a_1b) = 0$$

$$\text{If } ab_1 - a_1b = 0 \text{ then } \frac{a}{a_1} = \frac{b}{b_1} = \frac{\sqrt[3]{a^3 + b^3}}{\sqrt[3]{a_1^3 + b_1^3}} = \frac{c}{c} = 1$$

(Law of proportion)

$\therefore a = a_1$ and $b = b_1$ which is impossible

$\therefore ab_1 - a_1b \neq 0$

$$\text{Hence, } ab + ab_1 + a_1b = 0 \text{ or } \frac{a}{a_1} + \frac{b}{b_1} = -1$$

8. Ans. 0.50

Sol. Limit is of the form 1^∞ , so we have

$$\lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{2/x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} - 1 \right) \cdot \frac{2}{x}$$

$$\text{Now } = \lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} - 1 \right) \cdot \frac{2}{x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x - 3}{3x} \right) \cdot 2$$

$$= \lim_{x \rightarrow 0} \left(\frac{a^x - 1}{x} + \frac{b^x - 1}{x} + \frac{c^x - 1}{x} \right) \cdot \frac{2}{3}$$

$$= \left\{ \lim_{x \rightarrow 0} \frac{a^x - 1}{x} + \lim_{x \rightarrow 0} \frac{b^x - 1}{x} + \lim_{x \rightarrow 0} \frac{c^x - 1}{x} \right\} \cdot \frac{2}{3}$$

$$\frac{2}{3} (\ln a + \ln b + \ln c) = (\ln abc) \cdot \frac{2}{3}$$

$$\therefore \text{Desired limit} = e^{\frac{2}{3} \ln abc} = e^{\ln(abc)^{2/3}} = (abc)^{2/3}$$

9. Ans. 0.30

$$\text{Sol. } \begin{vmatrix} 1 & 1 & -2 \\ 2 & -1 & 1 \\ p & q & -r \end{vmatrix} = 0 \Rightarrow p + 5q - 3r = 0$$

$$\text{so, } 2px + 3qy + \frac{4}{3}(p + 5q) = 0$$

$$p \left(2x + \frac{4}{3} \right) + q \left(3y + \frac{20}{3} \right) = 0$$

$$\text{so fixed point is } \left(-\frac{2}{3}, -\frac{20}{9} \right)$$

$$\text{so the slope of required line} = -\frac{3}{10}$$

10. Ans. 2.00

$$\text{Sol. } f(x) = \frac{1}{(\{x\} + 1)^2}$$

$$\int_{-3/2}^{5/2} f(x) dx = 4 \int_0^1 \frac{1}{(\{x\} + 1)^2} dx$$